

# Solvers Mathematics (Class 7) — Summative Assessment 1 (SA1)

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## Official Question Paper & Student Workbook: Chapters 7 & 8

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**Curriculum Standard:** Solvers Mathematics-7 (NEP & NCERT Aligned)

**Portion:** Chapter 7 (A Tale of Three Intersecting Lines) & Chapter 8 (Working with Fractions)

**Instructions:**

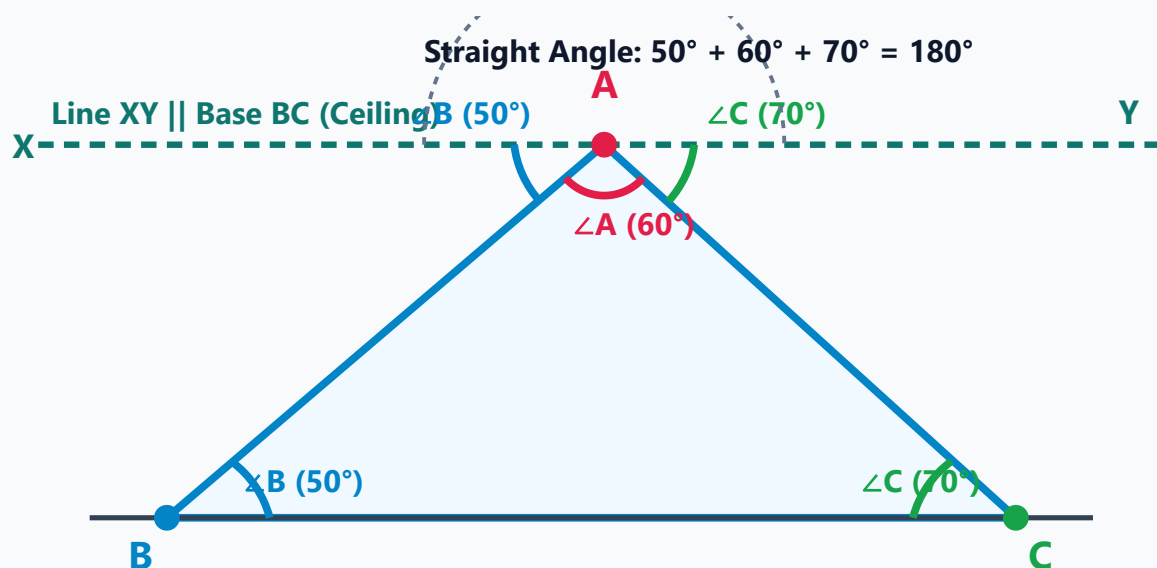
- 1. All questions are compulsory.
  - 2. Show complete rough work and calculations in the space provided.
  - 3. Draw neat geometric figures using a pencil, ruler, compass, and protractor wherever required.
  - 4. Certified Zero-Omission Coverage: Every single question from Exercises 7.1–7.6, Exercises 8.1–8.4, and Chapter Drills is included.
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## PART I: CHAPTER 7 — A TALE OF THREE INTERSECTING LINES (TRIANGLES)

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 **Visual Foundations & Diagrammatic Analysis**

Figure 7.1: Parallel Ceiling Model (Angle Sum Verification)



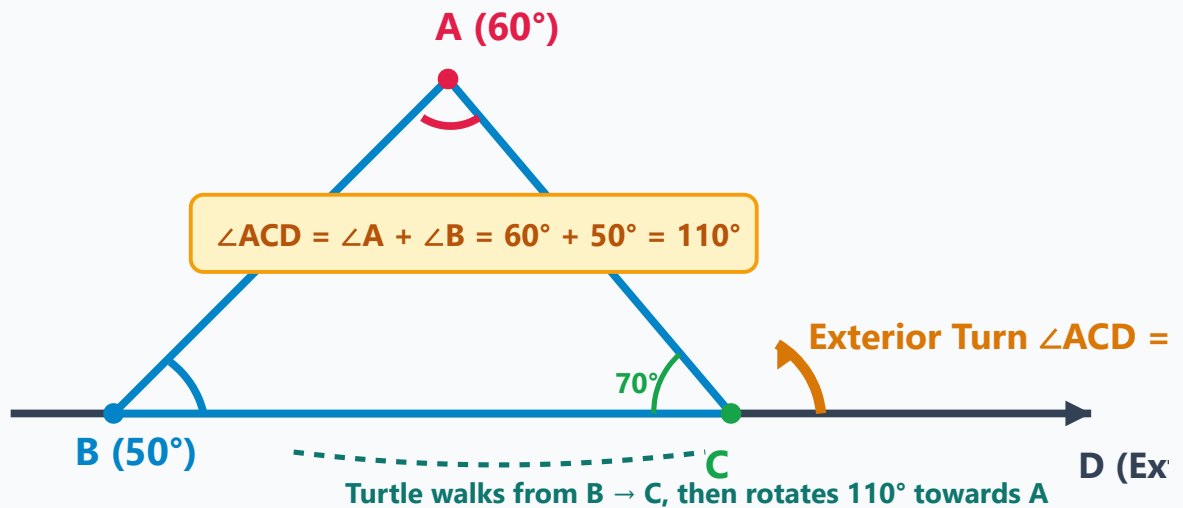
**Figure 7.1: Grant Sanderson's Parallel Ceiling Proof of the 180° Theorem**

**Geometric Intuition:** Do not treat  $180^\circ$  as an arbitrary decree. Draw a parallel ceiling line  $XY$  through apex  $A$ . The alternate interior 'Z-angles' effortlessly lift base angle  $B$  and base angle  $C$  to meet vertex angle  $A$ . The three angles unroll along the flat line into a single straight angle:

$$\angle A + \angle B + \angle C = 180^\circ!$$

\*Student Observation Task 1:\* In the diagram above, observe the parallel ceiling line  $XY \parallel BC$ . State the relationship between  $\angle XAB$  and  $\angle B$ , and explain why the three angles at vertex  $A$  must add up to a straight angle ( $180^\circ$ ). [2]

**Figure 7.2: Walking Turtle Exterior Angle Model**

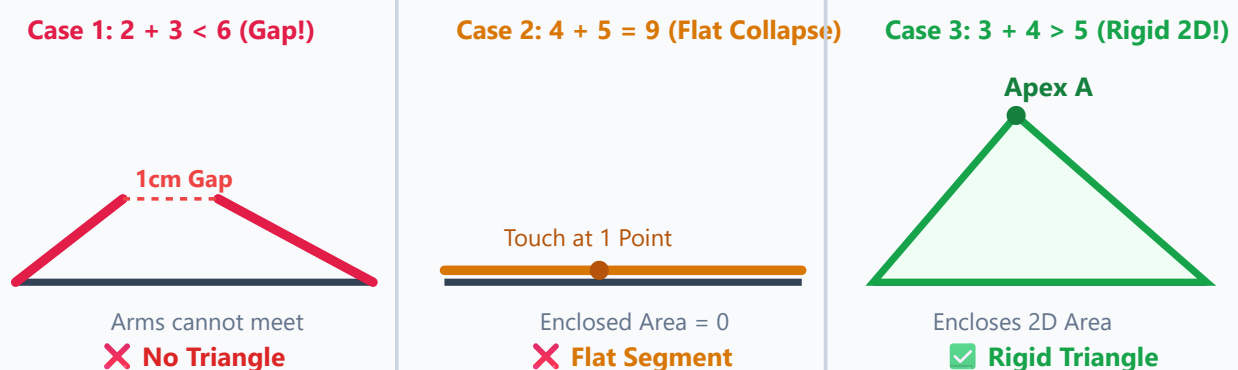


**Figure 7.2: The Walking Turtle Model (Exterior Angle Theorem)**

**Geometric Intuition:** Picture an orienting turtle navigating the perimeter. Walking from  $B$  through  $C$ , turning around to head back would be a straight angle ( $180^\circ$ ). Because the interior corner takes up  $70^\circ$ , the turtle must pivot through  $180^\circ - 70^\circ = 110^\circ$ . This rotational deficit is exactly supplied by the sum of the two remote interior turns:  $\angle_{\text{ext}} = \angle A + \angle B$ .

**\*Student Observation Task 2:\*** Trace the path of the walking turtle moving from  $B \rightarrow C$  and making a rotation to face vertex  $A$ . Write down the formula expressing exterior  $\angle ACD$  in terms of the remote interior angles  $\angle A$  and  $\angle B$ . [2]

**Figure 7.3: The Collapsing Drawbridge Model (Triangle Inequality Theorem)**

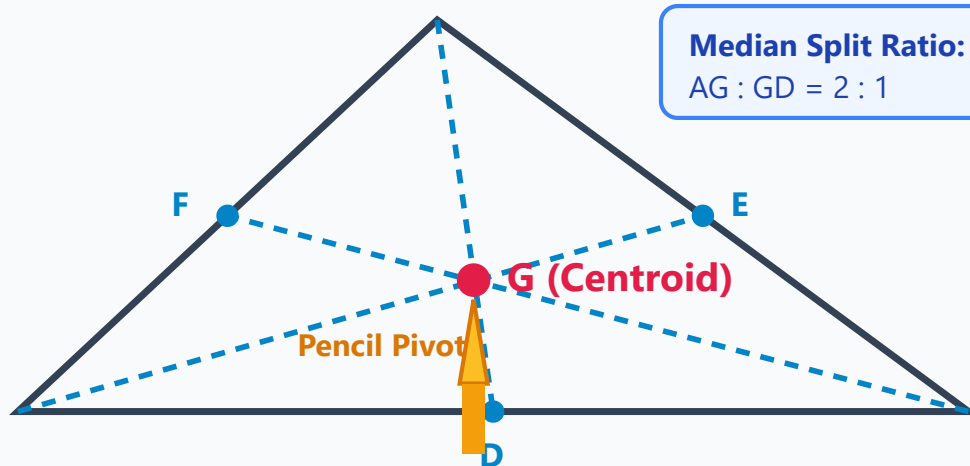


**Figure 7.3: The Collapsing Drawbridge (Triangle Inequality Theorem)**

**Geometric Intuition:** Picture two drawbridge arms pinned at opposite banks of a river of length  $c$ . If their combined reach is less than or equal to  $c$ , the arms collapse into the water or lie flat on the ground. A triangle can pop up into the third dimension of height **if and only if**  $a + b > c$ .

\*Student Observation Task 3:\* Explain using the three drawbridge cases shown above why three segment lengths of 2 cm, 3 cm, and 6 cm cannot form a triangle. [2]

**Figure 7.4: Centroid Balancing Center of Mass**



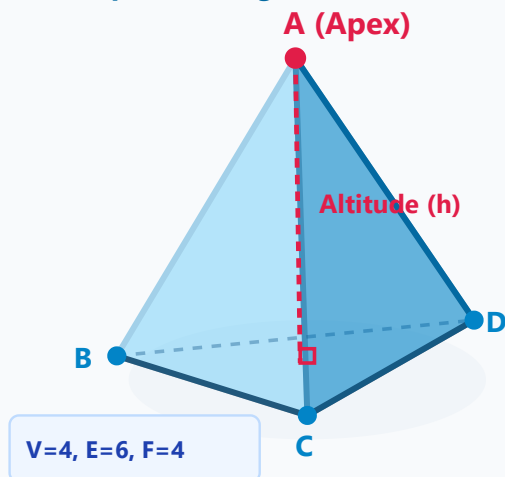
**Figure 7.4: The Physical Balancing Centroid (G) & 2:1 Ratio**

**Geometric Intuition:** Each median divides a triangle into two sub-triangles of identical physical area. All three medians intersect at one singular point: the **Centroid** ( $G$ ). You can balance any cardboard cut-out triangle horizontally on the sharp point of a pencil placed exactly at  $G$ !

\*Student Observation Task 4:\* Identify the point  $G$  in the triangle. In what ratio does  $G$  divide each median, and what physical property does  $G$  represent when balancing a cardboard triangle? [2]

**Figure 7.5: 3D Polyhedra & Dynamic Net Unfolding (Tetrahedron & Euler's Law)**

### 3D Perspective: Regular Tetrahedron



### Unfolded 2D Net (Euler: $V - E + F = 2$ )

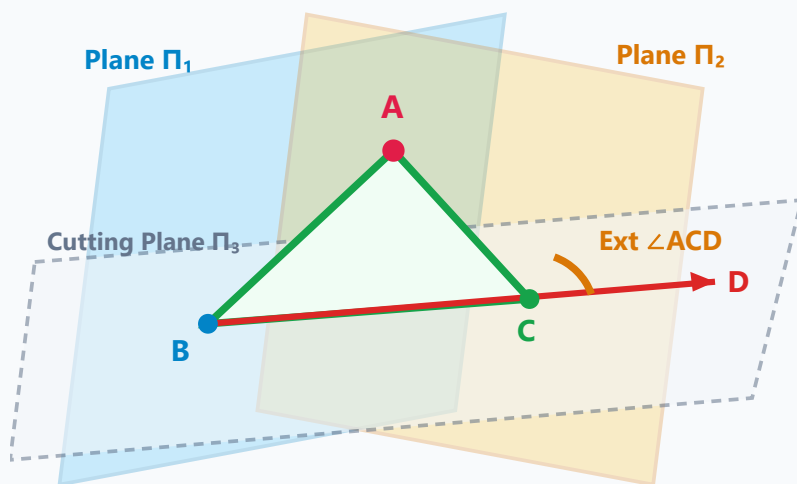
**Figure 7.5: 3D Polyhedra & Dynamic Net Unfolding (Tetrahedron & Euler's Law)**

**Spatial Intuition:** A triangle is not just a flat 2D area; it is the rigid architectural tile of 3D space. When 4 congruent equilateral triangles are joined, they form the regular tetrahedron (triangular pyramid). Unfolding the 3D solid yields a 2D net. Folding the flaps upwards unifies vertices  $A_1, A_2, A_3$  into a single 3D apex  $A$ . Every convex polyhedron strictly preserves Euler's invariant:  $V - E + F = 2$ .

\*Student Observation Task 5 (3D Polyhedra):\* Examine the 3D regular tetrahedron and its unfolded 2D net above. Verify Euler's polyhedral formula ( $V - E + F$ ) by counting its vertices, edges, and faces, and state how many triangular faces meet at each vertex. [2]

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### Figure 7.6: Three Non-Parallel Planes Slicing 3D Space



#### Spatial Angle Invariant:

- 3 Planes intersect along 3 lines
- 3 Lines trap a flat 2D Triangle
- Ray CD extends into 3D Space

$$\angle ACD = \angle A + \angle B \text{ in 3D!}$$

**Figure 7.6: Three Non-Parallel Planes Slicing 3D Space (Spatial Angle Geometry)**

**Spatial Intuition:** In 3D space, when three planes meet, each pair produces a straight line of intersection. These three lines trap a triangular cross-section on their shared plane. Extending a side along its spatial ray forms the 3D exterior angle. Regardless of how the triangle tilts or spins through 3D space, the planar exterior rotation law remains an invariant anchor:  $\angle ACD = \angle A + \angle B$ .

\*Student Observation Task 6 (3D Spatial Angles):\* In the 3D plane intersection diagram above, three non-parallel planes intersect to trap  $\triangle ABC$ . If interior  $\angle A = 55^\circ$  and interior  $\angle B = 65^\circ$ , calculate the measure of the 3D exterior angle  $\angle ACD$  along the intersection ray. [2]

## Mental Maths (Page 121)

1. Can a triangle have two right angles? Explain why or why not. [1]

2. Can a triangle have two obtuse angles? Give a reason. [1]

3. Can an equilateral triangle be obtuse-angled? State why. [1]

## Exercise 7.1: SSS Construction & Equilateral Triangles (Page 122)

1. Construct an equilateral triangle  $\triangle ABC$  with each side measuring 5 cm using a ruler and compass. Verify your construction by measuring each angle with a protractor. [3]

2. Draw an equilateral triangle  $\triangle XYZ$  with each side measuring 5.5 cm. [2]

3. Construct an isosceles triangle  $\triangle PQR$  in which base  $QR = 5$  cm and the two equal sides  $PQ$  and  $PR$  each measure 7 cm. [3]
  4. Construct a triangle  $\triangle PQR$  with side lengths  $PQ = 7$  cm,  $QR = 8$  cm, and  $PR = 6$  cm. [3]
  5. Draw a triangle with sides 4 cm, 6 cm, and 7 cm, and label its vertices as  $L$ ,  $M$ ,  $N$ . [2]
  6. Kunal is preparing a triangular wooden frame for a science exhibition. The three sides of the frame measure 45 cm, 55 cm, and 70 cm. Verify whether it is possible to construct this wooden frame. [2]
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## Exercise 7.2: Triangle Inequality Property (Pages 123–125)

1. Can a triangle have side lengths 5 cm, 7 cm, and 14 cm? Justify your answer. [2]
  2. Determine which of the following sets of side lengths can form a valid triangle. Give reasons for each: [3]
    - (a) 2, 2, 5
    - (b) 3, 4, 6
    - (c) 4, 5, 8
    - (d) 10, 20, 25
    - (e) 15, 20, 35
    - (f) 24, 26, 28
  3. The lengths of two sides of a triangle are 7 cm and 11 cm. Between which two values must the length of the third side fall? [2]
  4. In triangle construction using circles, explain what geometric condition occurs when the sum of the radii of two circles drawn at the ends of a base segment is exactly equal to the base length. [2]
- 

## Exercise 7.3: Side-Angle-Side (SAS) Construction (Page 126)

1. Construct a triangle  $\triangle ABC$  such that the length of side  $AB = 6.4$  cm, the length of side  $BC = 4.8$  cm, and the included angle  $\angle B = 75^\circ$ . [3]
  2. Construct a triangle  $\triangle PQR$  such that  $PQ = 6.5$  cm,  $PR = 6.5$  cm, and  $\angle P = 115^\circ$ . After constructing the triangle, identify the type of triangle formed based on: [3]
    - (a) The lengths of its sides
    - (b) The measures of its angles
  3. Construct an isosceles triangle in which each of the equal sides is 6 cm and the angle included between them is  $50^\circ$ . [3]
  4. Construct triangles for the following measurements where the angle is included between the two sides: [3]
    - (a) 3 cm,  $75^\circ$ , 7 cm
    - (b) 6 cm,  $25^\circ$ , 3 cm
    - (c) 3 cm,  $120^\circ$ , 8 cm
-

### Exercise 7.4: Angle-Side-Angle (ASA) Construction (Page 127)

1. Construct a triangle  $\triangle ABC$  such that base  $BC = 5.5$  cm, the angle at point  $B$  is  $60^\circ$ , and the angle at point  $C$  is  $45^\circ$ . [3]
2. Draw a triangle  $\triangle XYZ$  where  $XY = 7$  cm,  $\angle X = 30^\circ$ , and  $\angle Y = 110^\circ$ . Determine whether it is an obtuse-angled triangle. [3]
3. Construct a triangle with a base length of 6 cm and both base angles measuring  $50^\circ$  each. Measure the remaining two sides of the triangle and record your observation. [3]
4. Construct triangles for the following measurements: [3]
  - (a)  $75^\circ$ , 5 cm,  $75^\circ$
  - (b)  $25^\circ$ , 3 cm,  $60^\circ$
  - (c)  $120^\circ$ , 6 cm,  $30^\circ$
5. Construct a right-angled triangle whose one leg is 6 cm and one of whose acute angles measures  $60^\circ$ . [3]

### Exercise 7.5: Angle Sum Property of a Triangle (Page 130)

1. Take three non-collinear points  $A$ ,  $B$ , and  $C$  on a page of your notebook. Join  $AB$ ,  $BC$ , and  $CA$ . What figure do you get? Name: [2]
  - (a) The side opposite to  $\angle B$
  - (b) The angle opposite to side  $AB$
  - (c) The vertex opposite to side  $BC$
  - (d) The side opposite to vertex  $B$
2. The three angles of a triangle are equal to one another. What is the measure of each angle? What kind of triangle is it? [2]
3. Find the third angle of a triangle when two of the angles are: [2]
  - (a)  $36^\circ$  and  $72^\circ$
  - (b)  $150^\circ$  and  $15^\circ$
  - (c)  $90^\circ$  and  $30^\circ$
  - (d)  $75^\circ$  and  $45^\circ$
4. The angles of a triangle are in the ratio 1 : 2 : 3. Determine the measures of all three angles. [2]
5. In a triangle, if two angles are  $50^\circ$  and  $70^\circ$ , find the measure of the third angle  $x$ . [2]
6. The angles of a triangle are arranged in ascending order such that the difference between consecutive angles is  $10^\circ$ . Find the three angles. [3]
7. If one angle of a triangle is  $60^\circ$  and the other two angles are in the ratio 1 : 5, find the angles. [3]
8. Two angles of a triangle are equal and the third angle measures  $30^\circ$ . Determine all the angles of the triangle. [2]
9. If one angle of a triangle is equal to the sum of the other two angles, prove that the triangle is a right triangle. [3]

10. The two acute angles of a right triangle differ by  $20^\circ$ . Find the two angles. [3]
11. In  $\triangle PQR$ ,  $\angle P = (2x)^\circ$ ,  $\angle Q = (3x)^\circ$ , and  $\angle R = (5x)^\circ$ . Find the value of  $x$  and all angles. [3]
12. What happens to the area and the figure when three vertices of a triangle lie on a single straight line? [1]

### Exercise 7.6: Exterior Angle Property & Altitudes (Pages 131–133)

1. In the given figure,  $\angle x : \angle y = 5 : 3$  and the exterior angle  $\angle ACD = 160^\circ$ . Find the values of  $x$ ,  $y$ , and  $z$  (where  $z = \angle ACB$ ). [3]
2. Compute the value of  $x$  in each of the following cases: [3]
- (a) Exterior angle is  $110^\circ$ , one interior opposite angle is  $50^\circ$ .
  - (b) Exterior angle is  $125^\circ$ , one interior opposite angle is  $65^\circ$ .
  - (c) Interior opposite angles are  $40^\circ$  and  $70^\circ$ , exterior angle is  $x$ .
3. An exterior angle of a triangle is  $115^\circ$  and one of the interior opposite angles is  $35^\circ$ . Find the other interior opposite angle. [2]
4. One of the exterior angles of a triangle is  $120^\circ$  and the interior opposite angles are in the ratio  $1 : 2$ . Find all the angles of the triangle. [3]
5. In  $\triangle ABC$ , side  $BC$  is produced to  $D$ . If  $\angle ACD = 105^\circ$  and  $\angle BAC = 45^\circ$ , find  $\angle ABC$  and  $\angle ACB$ . [3]
6. In  $\triangle ABC$ ,  $AC \perp BD$ , and the angles are in ratio  $\angle A : \angle B : \angle C = 3 : 2 : 1$ . Find the angles. [3]
7. Describe the location of the orthocenter in: [2]
- (a) An acute-angled triangle
  - (b) A right-angled triangle
  - (c) An obtuse-angled triangle
8.  $\triangle ABC$  is an isosceles triangle with  $AB = AC$ . Ray  $AX$  bisects the exterior angle  $\angle DAC$ . If  $\angle B = 70^\circ$ , find  $\angle ACB$  and prove that  $AX \parallel BC$ . [4]

### Chapter 7 Drill (Pages 135–138)

#### Section A: Multiple Choice Questions (1 Mark each)

1. In triangle construction, if the sum of two sides is exactly equal to the third side, the arcs drawn will:
- (a) Touch at one point
  - (b) Intersect at two points
  - (c) Intersect internally
  - (d) Not meet at all
2. Which set of lengths forms a valid triangle?
- (a) 2, 3, 6
  - (b) 4, 5, 9

(c) 3, 4, 5

(d) 1, 2, 4

3. The altitude of a right-angled triangle drawn from the right angle vertex:

(a) Lies outside

(b) Coincides with hypotenuse

(c) Lies inside the triangle

(d) Is not possible

4. If two sides of a triangle are 5 cm and 6 cm, which of the following CANNOT be the third side?

(a) 12 cm

(b) 4 cm

(c) 8 cm

(d) 2 cm

5. In a right-angled triangle, the longest side is called the:

(a) Median

(b) Hypotenuse

(c) Altitude

(d) Base

6. The point where the three medians of a triangle meet is the:

(a) Midpoint

(b) Vertex

(c) Centroid

(d) Orthocenter

7. An altitude is always:

(a) Parallel to base

(b) Perpendicular to base

(c) Equal to base

(d) Curved

8. A triangle cannot exist if:

(a) One side = sum of other two

(b) All sides equal

(c) All angles acute

(d) One angle is right

9. To construct a unique triangle using the SAS criterion, the angle must be:

(a) Acute

(b) Right angle

(c) Included between the given sides

(d) Any angle

10. If an exterior angle of a triangle measures  $110^\circ$  and one of its interior opposite angles is  $50^\circ$ , what is the measure of the other interior opposite angle?

- (a)  $50^\circ$
- (b)  $60^\circ$
- (c)  $70^\circ$
- (d)  $180^\circ$

### Section B: Fill in the Blanks (1 Mark each)

1. A triangle is formed by joining three \_\_\_\_\_ points.
2. A perpendicular drawn from a vertex to the opposite side is called an \_\_\_\_\_.
3. A triangle in which one angle is  $90^\circ$  is called a \_\_\_\_\_ triangle.
4. In an \_\_\_\_\_ triangle, at least two sides are equal.
5. Triangle inequality states that each side must be \_\_\_\_\_ than the sum of the other two sides.
6. The intersection point of altitudes is called the \_\_\_\_\_.

### Section C: Very Short Answer Questions (1 Mark each)

1. Can a triangle have angles measuring  $100^\circ$ ,  $50^\circ$ , and  $40^\circ$ ? Justify. [1]
2. Define the median of a triangle. [1]
3. What is the measure of each angle of an equilateral triangle? [1]
4. Does an equilateral triangle with sides 50 cm, 50 cm, 50 cm exist? In general, does an equilateral triangle exist for any given side length? [1]
5. Which side is the longest side in a right-angled triangle? [1]
6. If the sides of a triangle are 2 cm, 3 cm, and 5 cm, is it possible to form a triangle? [1]
7. How many vertices does a triangle have? [1]
8. Find the third angle of a triangle if the other two angles are  $58^\circ$  and  $62^\circ$ . [1]
9. Find the exterior angle of a triangle if the interior opposite angles are  $46^\circ$  and  $54^\circ$ . [1]

### Section D: Short Answer Questions (2 Marks each)

1. Two angles of a triangle measure  $45^\circ$  and  $65^\circ$ . Find the measure of the third angle. [2]
2. The angles of a triangle are  $(x - 40)^\circ$ ,  $(x - 20)^\circ$ , and  $(\frac{1}{2}x - 10)^\circ$ . Find the value of  $x$ . [2]
3. An isosceles triangle has a vertex angle of  $50^\circ$ . Find the measures of its base angles. [2]
4. Why can a triangle not have two right angles? [2]
5. If the perimeter of an equilateral triangle is 36 cm, find the length of each side. [2]
6. In a triangle, an exterior angle measures  $130^\circ$  and one interior opposite angle is  $70^\circ$ . Find the other interior opposite angle. [2]

### Section E: Long Answer Questions (4 Marks each)

1. State and prove the **Angle Sum Property of a Triangle** using a parallel auxiliary line. [4]
2. In  $\triangle ABC$ ,  $\angle B = 2\angle A$  and  $\angle C = 3\angle A$ . Find all three angles and classify the triangle. [4]

3. Construct  $\triangle ABC$  where side  $AB = 6$  cm,  $\angle A = 60^\circ$ , and  $\angle B = 45^\circ$  using a ruler and compass. Calculate the measure of the third angle  $\angle C$ . [4]
4. An exterior angle of a triangle is  $120^\circ$  and one of the interior opposite angles is  $40^\circ$ . Find all the interior angles of the triangle. [4]
5. In  $\triangle DEF$ , the lengths of sides are  $DE = 4.5$  cm,  $EF = 5.5$  cm, and  $DF = 10$  cm. Can this triangle be constructed? Justify using the triangle inequality theorem. [4]
6. In  $\triangle ABC$ ,  $AD \perp BC$  and  $BE \perp AC$ . If  $\angle BAC = 50^\circ$  and  $\angle ABC = 70^\circ$ , find  $\angle BCA$ ,  $\angle CAD$ , and  $\angle CBE$ . [4]

### Section F: Competency-Based HOTS & Life Skills (Page 138)

1. Three sticks measure 10 cm, 14 cm, and 25 cm. If the longest stick is broken into two pieces of 12 cm and 13 cm, determine if a triangle can be formed with the original two sticks and one of the broken pieces. [3]
2. If the sides of a triangle are  $(x + 1)$ ,  $(x + 2)$ , and 5, find the complete mathematical range of possible values for  $x$ . [3]
3. Explain how the structural rigidity of triangles in engineering bridges and roof trusses reflects the importance of core moral values in personal character. [2]

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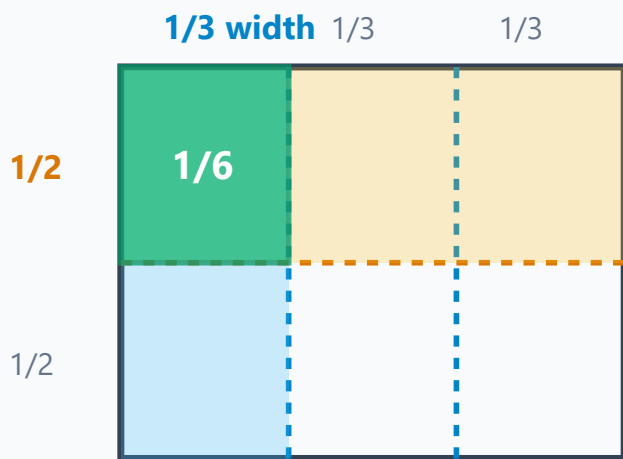
## PART II: CHAPTER 8 — WORKING WITH FRACTIONS

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### Visual Foundations & Diagrammatic Analysis

Figure 8.1: 2D Area Dissection Model for Fraction Multiplication



### Area Multiplication Rule:

$$\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

- Numerators:  $1 \times 1 = 1$  tile
- Denominators:  $2 \times 3 = 6$  total

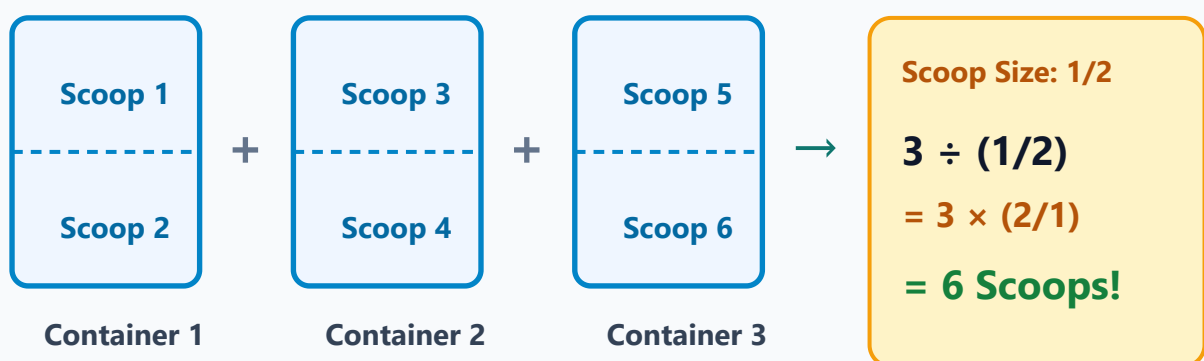
**Area of sub-rectangle!**

**Figure 8.1: The 2D Area Dissection Model for Fraction Multiplication**

**Visual Intuition:** Why is  $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ ? Slice a  $1 \times 1$  unit square vertically into 3 columns (each  $\frac{1}{3}$  wide). Slice horizontally into 2 rows (each  $\frac{1}{2}$  tall). The double-shaded rectangular cell where the top row overlaps the left column represents 'half of one-third'. The whole square now contains  $2 \times 3 = 6$  identical small tiles, and exactly 1 tile is selected!

\*Student Observation Task 1:\* In the unit square model above, explain how shading  $\frac{1}{3}$  of the width and  $\frac{1}{2}$  of the height visually proves that  $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ . [2]

**Figure 8.2: Measuring Scoop Division Model**

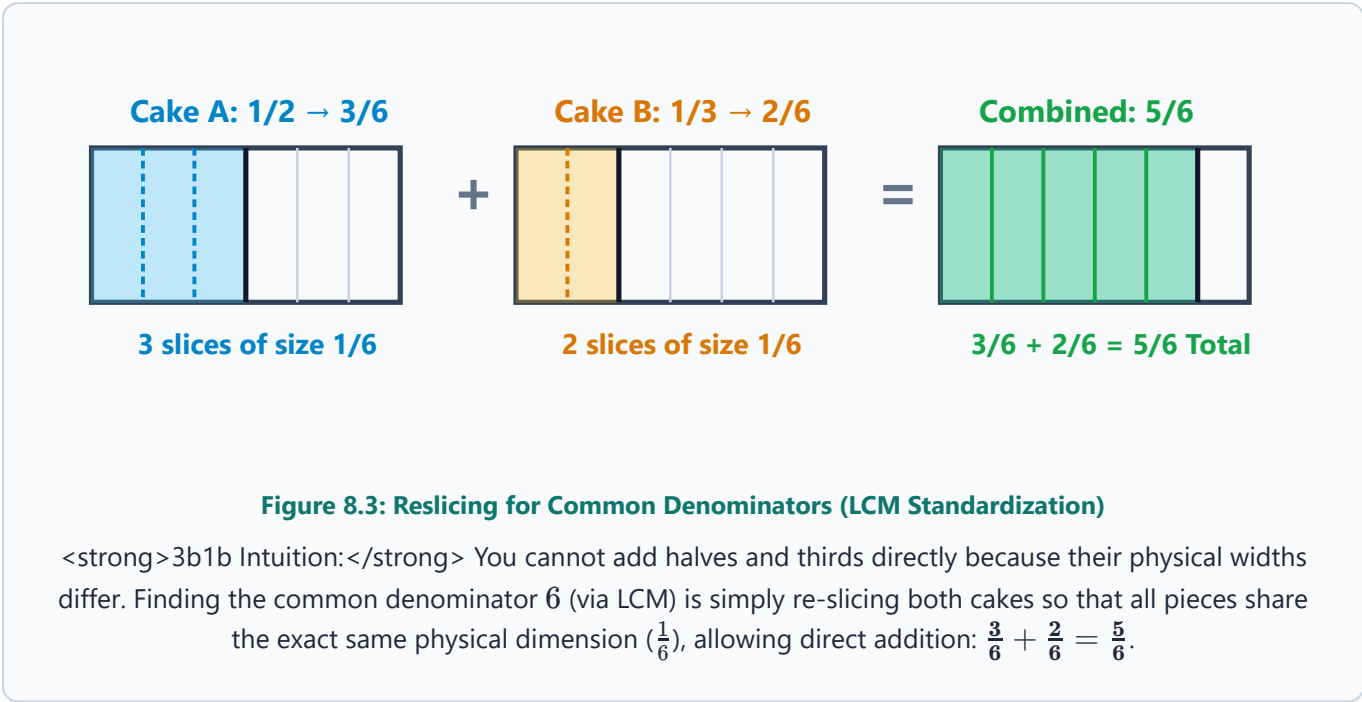


**Figure 8.2: The Measuring Scoop Model for Fraction Division**

**Physical Intuition:** Why do we 'flip and multiply' when dividing by  $\frac{1}{2}$ ? Translate  $3 \div \frac{1}{2}$  into reality: 'How many  $\frac{1}{2}$ -cup scoops of sugar fit in a 3-cup bag?' Every 1 whole cup holds 2 scoops (the reciprocal  $1 \div \frac{1}{2} = 2$ ). Therefore, 3 whole cups must hold  $3 \times 2 = 6$  scoops!

\*Student Observation Task 2:\* State the physical question represented by  $3 \div \frac{1}{2}$  using measuring scoops, and explain why division results in multiplying by the reciprocal 2. [2]

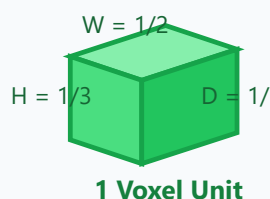
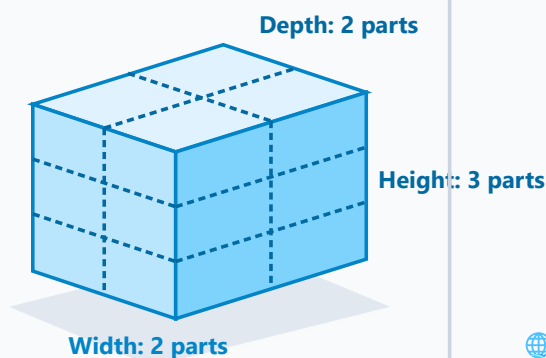
Figure 8.3: Common Denominator Reslicing



\*Student Observation Task 3:\* Illustrate why  $\frac{1}{2}$  and  $\frac{1}{3}$  cannot be added directly without reslicing both rectangles into sixths ( $\frac{1}{6}$ ). [2]

Figure 8.4: 3D Volumetric Fraction Voxel Slicer

**Unit Cube ( $1 \times 1 \times 1$ ) Sliced into 12 Voxels** Exploded 3D Voxel:  $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} = \frac{1}{12}$



**Volume Formula:**

$$V = W \times H \times D$$

$$= \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2}$$

$$= \frac{1}{12} \text{ Total}$$

12 voxels fill 1 whole cube

**Interactive 3D: See & explode this in maths\_sa1.**

**Figure 8.4: 3D Volumetric Fraction Voxel Slicer (Dissecting Unit Volume)**

**Spatial Intuition:** Multiplication extends from numbers (1D lines) to area (2D tiles) to volume (3D space). Slicing a unit cube into 2 parts along width, 3 parts along height, and 2 parts along depth partitions 3D space into  $2 \times 3 \times 2 = 12$  identical sub-cubes (voxels). Each voxel has dimensions  $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} = \frac{1}{12}$  of the entire volume!

**\*Student Observation Task 4 (3D Volumetric Fractions):\*** In the 3D unit cube model above, the cube is partitioned into 2 slices along width, 3 slices along height, and 2 slices along depth. Calculate the total number of voxel sub-cubes created, and express the volume of a single voxel as a fraction of the total unit cube volume. [2]

## Exercise 8.1: Taxonomy of Fractions & Equivalent Fractions (Page 141)

1. Classify each of the following as a proper fraction, improper fraction, or mixed fraction: [2]

$$\frac{3}{5}, \quad \frac{7}{4}, \quad 2\frac{1}{3}, \quad \frac{1}{8}, \quad \frac{11}{11}, \quad \frac{15}{2}, \quad 4\frac{3}{7}, \quad \frac{9}{10}$$

2. Convert the following mixed fractions into improper fractions: [2]

(a)  $2\frac{1}{3}$

(b)  $3\frac{2}{5}$

(c)  $7\frac{1}{2}$

(d)  $5\frac{3}{4}$

(e)  $2\frac{5}{8}$

3. Find the missing numbers that make each set of fractions equal: [2]

(a)  $\frac{3}{5} = \frac{\square}{20} = \frac{18}{\square}$

(b)  $\frac{4}{7} = \frac{28}{\square} = \frac{\square}{49}$

(c)  $\frac{5}{8} = \frac{15}{\square} = \frac{\square}{64}$

(d)  $\frac{6}{11} = \frac{\square}{33} = \frac{42}{\square}$

4. Write any three equivalent fractions for each of the following: [2]

(a)  $\frac{2}{3}$

(b)  $\frac{4}{5}$

(c)  $\frac{5}{6}$

5. From each group given below, identify the set of like fractions: [2]

(a)  $\frac{2}{7}, \frac{5}{7}, \frac{9}{7}, \frac{3}{7}$

(b)  $\frac{3}{4}, \frac{3}{5}, \frac{3}{8}, \frac{3}{11}$

(c)  $\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}$

6. Check whether the fractions  $\frac{4}{6}$  and  $\frac{20}{30}$  are equivalent or not. [2]

7. Aarav ate  $\frac{7}{4}$  of a pizza. Write this fraction as a mixed fraction and state whether Aarav ate more than one full pizza or less than one full pizza. [2]

8. Which unit fraction is smaller:  $\frac{1}{9}$  or  $\frac{1}{15}$ ? Give a reason for your answer. [1]

9. A tailor has  $\frac{3}{4}$  metre of silk cloth. He wants to write this amount as an equivalent fraction with a denominator of 12 to mark his measurements. What will be the new numerator? [2]

10. Sana planned to read  $\frac{3}{5}$  of her book in a day. By evening, she had read  $\frac{12}{20}$  of the book. Did Sana achieve her reading goal? Explain your answer using the concept of equivalent fractions. [2]

## Exercise 8.2: Comparing, Adding & Subtracting Unlike Fractions (Pages 143–144)

1. Fill in the blanks using the correct sign ( $>$ ,  $<$ , or  $=$ ): [2]

(a)  $\frac{5}{7} \square \frac{3}{7}$

(b)  $\frac{1}{4} \square \frac{1}{6}$

(c)  $\frac{3}{8} \square \frac{5}{12}$

(d)  $\frac{7}{9} \square \frac{14}{18}$

2. Insert the correct sign ( $<$ ,  $>$ ,  $=$ ): [2]

(a)  $\frac{2}{3} \square \frac{5}{6}$

(b)  $\frac{7}{10} \square \frac{4}{5}$

(c)  $\frac{5}{12} \square \frac{7}{16}$

3. Arrange the following fractions in ascending order: [3]

(a)  $\frac{2}{3}, \frac{5}{6}, \frac{7}{9}, \frac{1}{2}$

(b)  $\frac{3}{5}, \frac{7}{10}, \frac{4}{15}, \frac{1}{2}$

4. Find the sum and write the answer as a mixed fraction: [3]

(a)  $3\frac{2}{5} + 2\frac{3}{4}$

(b)  $4\frac{1}{6} + 3\frac{5}{12}$

(c)  $2\frac{3}{8} + 1\frac{5}{6}$

5. Subtract the following: [3]

(a)  $\frac{5}{6} - \frac{2}{5}$

(b)  $12 - 5\frac{4}{9}$

(c)  $8 - 3\frac{5}{6}$

6. Evaluate: [3]

(a)  $3\frac{1}{2} - 1\frac{5}{6} + 2\frac{2}{3}$

(b)  $2\frac{3}{4} - \frac{1}{2} + 1\frac{1}{8}$

7. A wire is  $15\frac{3}{5}$  metres long. If a piece of  $6\frac{2}{3}$  metres is cut off, find the length of the wire left. [3]

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### Exercise 8.3: Multiplication of Fractions & Lowest Form (Page 146)

1. Multiply and simplify to lowest form: [2]

(a)  $\frac{2}{3} \times \frac{5}{4}$

(b)  $\frac{3}{8} \times \frac{4}{9}$

(c)  $2\frac{1}{3} \times 1\frac{1}{5}$

(d)  $\frac{5}{12} \times \frac{9}{10}$

2. Convert into mixed fractions: [2]

(a)  $7 \times \frac{2}{3}$

(b)  $4 \times \frac{3}{5}$

(c)  $5 \times \frac{4}{7}$

3. Find the area of a rectangular playground that is  $14\frac{2}{5}$  metres long and  $9\frac{5}{8}$  metres wide. [3]

4. Evaluate: [3]

(a)  $\frac{3}{8}$  of 40

(b)  $\frac{2}{3}$  of 1 hour (in minutes)

(c)  $\frac{3}{4}$  of 2 kilograms (in grams)

5. A car runs 16 km using 1 litre of petrol. How much distance will it cover using  $2\frac{3}{4}$  litres of petrol? [3]

6. Rohan has a novel containing 120 pages. He reads  $\frac{3}{4}$  of the book. [3]

(a) What fraction of the book has he not read?

(b) How many pages has he read?

(c) How many pages remain to be read?

7. A water tanker supplies  $32\frac{1}{2}$  kilolitres of water in a day.  $\frac{2}{5}$  of this water is used for household purposes. Find the amount of water used for household purposes. [3]

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### Exercise 8.4: Division of Fractions & Reciprocals (Pages 147–148)

1. Write the reciprocal of each of the following: [2]

- (a)  $\frac{7}{4}$
- (b)  $\frac{1}{9}$
- (c)  $2\frac{1}{3}$
- (d) 6
- (e)  $\frac{5}{8}$

2. Divide: [3]

- (a)  $4 \div \frac{1}{2}$
- (b)  $\frac{3}{5} \div 6$
- (c)  $5\frac{1}{4} \div 2\frac{1}{3}$
- (d)  $\frac{7}{9} \div \frac{14}{27}$
- (e)  $9 \div \frac{3}{4}$

3. A rope of length  $15\frac{3}{4}$  metres is cut into equal pieces of length  $\frac{3}{4}$  metre each. Find the number of pieces. [3]

4. A shopkeeper has 27 kg of dry fruits. He distributes them equally among 36 children. How many grams of dry fruits does each child receive? [3]

5. A garden plot is divided into 4 equal sections.  $\frac{2}{3}$  of each section is planted with flowers. What fraction of the entire garden is planted with flowers? If the total area of the garden is  $240 \text{ m}^2$ , find the planted area. [3]

6. The cost of  $7\frac{1}{2}$  kg of sugar is ₹340. Find the price of 1 kg of sugar. [3]

7. A wire of length  $18\frac{3}{4}$  metres is distributed equally among 15 students for a science project. Find the length of wire received by each student. [3]

8. A tailor uses  $2\frac{1}{4}$  metres of cloth to stitch one shirt. How many such shirts can be stitched from a roll of  $33\frac{3}{4}$  metres? [3]

9. Mira is reading a novel that has 480 pages. She read  $\frac{5}{8}$  of the pages today. How many more pages does she need to read to complete the book? [3]

10. A car travels a distance of  $202\frac{1}{2}$  km in  $4\frac{1}{2}$  hours. What is the average speed of the car in km/h? [3]

## Chapter 8 Drill (Pages 150–153)

### Section A: Multiple Choice Questions (1 Mark each)

1. Which of the following is a unit fraction?

- (a)  $\frac{4}{1}$
- (b)  $\frac{1}{7}$
- (c)  $\frac{7}{4}$
- (d)  $\frac{4}{7}$

2. The reciprocal of  $2\frac{1}{3}$  is:

- (a)  $\frac{7}{3}$
- (b)  $\frac{3}{7}$
- (c)  $\frac{1}{7}$

(d)  $2\frac{3}{1}$

3. To compare  $\frac{2}{5}$  and  $\frac{3}{4}$ , the common denominator should be:

(a) 5

(b) 4

(c) 20

(d) 10

4. A fraction is in its simplest form if its numerator and denominator are:

(a) Equal

(b) Co-prime

(c) Prime numbers

(d) Even numbers

5. What is  $\frac{3}{4}$  of 16?

(a) 12

(b) 8

(c) 4

(d) 16

6. Which symbol makes the statement true:  $\frac{3}{4} \square \frac{9}{12}$ ?

(a)  $>$

(b)  $<$

(c)  $=$

(d)  $\leq$

7. An improper fraction is always:

(a) Less than 1

(b) Greater than or equal to 1

(c) Equal to 1

(d) A unit fraction

8.  $4 \div \frac{1}{2}$  is equal to:

(a) 2

(b) 8

(c)  $\frac{1}{8}$

(d) 4

9. An equivalent fraction of  $\frac{2}{3}$  with denominator 12 is:

(a)  $\frac{4}{12}$

(b)  $\frac{8}{12}$

(c)  $\frac{6}{12}$

(d)  $\frac{10}{12}$

10. If  $\frac{5}{8} = \frac{15}{x}$ , then the value of  $x$  is:

- (a) 16
- (b) 24
- (c) 30
- (d) 40

### Section B: Fill in the Blanks (1 Mark each)

- The number that does not have a reciprocal is \_\_\_\_\_.
- In the fraction  $\frac{7}{11}$ , the top number 7 is called the \_\_\_\_\_.
- Fractions with the same denominator are called \_\_\_\_\_ fractions.
- While dividing fractions, we multiply by the \_\_\_\_\_ of the divisor.
- The product of  $\frac{3}{5} \times \frac{5}{3}$  is \_\_\_\_\_.
- A fraction written as  $3\frac{1}{2}$  is called a \_\_\_\_\_ fraction.

### Section C: Very Short Answer Questions (1 Mark each)

- Define a proper fraction and give one example. [1]
- Convert  $\frac{22}{4}$  into a mixed fraction. [1]
- Find the HCF of 24 and 36, and use it to simplify  $\frac{24}{36}$ . [1]
- Write the reciprocal of  $3\frac{1}{5}$ . [1]
- Is  $\frac{12}{12}$  a proper fraction or an improper fraction? Give a reason. [1]
- Multiply  $9 \times \frac{4}{27}$ . [1]
- Write the next two equivalent fractions for  $\frac{3}{7}$ . [1]
- Compare  $\frac{5}{8}$  and  $\frac{7}{12}$  using any suitable method. [1]
- How many  $\frac{2}{3}$  kg pieces can be cut from a cake of weight 4 kg? [1]
- Divide  $\frac{5}{8}$  by  $(\frac{3}{4} \text{ of } \frac{5}{6})$ . [1]
- Which is heavier:  $\frac{12}{25}$  of 500 grams or  $\frac{3}{20}$  of 4 kg? [1]

### Section D: Short Answer Questions (2 Marks each)

- Simplify  $\frac{72}{108}$  using the cancellation method, showing all steps. [2]
- Arrange the fractions  $\frac{2}{3}$ ,  $\frac{5}{6}$ ,  $\frac{7}{12}$  in ascending order. [2]
- A rope is  $\frac{3}{4}$  metre long. How many such ropes are required to make 6 metres? [2]
- Find the value of  $x$  if  $\frac{x}{7} = \frac{15}{35}$ . [2]
- Convert the mixed fraction  $7\frac{3}{8}$  into an improper fraction and then find its reciprocal. [2]
- Solve:  $(2\frac{1}{2} \times 1\frac{1}{5}) \div \frac{3}{4}$ . [2]

### Section E: Long Answer Questions (4 Marks each)

- The area of a rectangular plot is  $65\frac{1}{3} \text{ m}^2$ . If its breadth is  $5\frac{1}{3}$  metres, find its length. [4]
- $18\frac{3}{4}$  litres of juice is poured into small bottles of capacity  $\frac{3}{4}$  litre each. How many bottles can be filled completely? [4]

3. Mili reads  $\frac{1}{3}$  part of a book in 1 hour. How much part of the book will she read in  $2\frac{1}{5}$  hours? [4]
4. Vidya and Pratap went for a picnic. Their mother gave them a water bottle that contained 5 litres of water. Vidya consumed  $\frac{2}{5}$  of the water. Pratap consumed the remaining water. [4]
- (a) How much water did Vidya drink?
- (b) What fraction of the total quantity of water did Pratap drink?
5. Saili plants 4 saplings in a row in her garden. The distance between two adjacent saplings is  $\frac{3}{4}$  metre. Find the distance between the first and the last sapling. [4]
6. In a class of 40 students,  $\frac{1}{5}$  of the total students like to study English,  $\frac{2}{5}$  like to study Mathematics, and the remaining students like to study Science. [4]
- (a) How many students like to study English?
- (b) How many students like to study Mathematics?
- (c) What fraction of the total students like to study Science?
7. A cement company earns a profit of ₹8 per bag of white cement sold and a loss of ₹5 per bag of grey cement sold. If it sells 3,000 bags of white cement and 5,000 bags of grey cement in a month, calculate its net profit or loss. [4]
8. Evaluate:  $\left(\frac{2}{3} + \frac{1}{2}\right) \times \left(\frac{5}{6} - \frac{1}{4}\right) \div \frac{7}{12}$ . [4]
9. A rectangular field is  $24\frac{1}{2}$  m long and 15 m broad. Find its perimeter and the total cost of fencing it at ₹20 per metre. [4]
10. A family consumes  $3\frac{1}{2}$  kg of rice in a week. How much rice will they consume in  $4\frac{2}{7}$  weeks? [4]

## Section F: HOTS & Life Skills (Page 153)

1. Shade  $\frac{1}{2}$  of a rectangle. Now, divide the shaded part into 3 equal parts and take one part. What fraction of the original rectangle is this? [2]
2. What is the reciprocal of  $\frac{1}{5}$ ? [1]
3. Which is greater:  $\frac{1}{2}$  of 20 or  $\frac{1}{4}$  of 40? [2]
4. If you have  $\frac{3}{4}$  of a pizza and share it equally among 3 people, what fraction of a whole pizza does each person receive? [2]
5. How does sharing resources (like fractional portions of food, time, or knowledge) build empathy and community cooperation? [2]

# PART III: TIMED MODEL ASSESSMENT TEST (40 MARKS)

**Time Allowed:** 1 Hour | **Maximum Marks:** 40 Marks

## SECTION A: OBJECTIVE TYPE QUESTIONS (10 Marks)

1. The sum of the interior angles of any triangle is always: [1]
- (a)  $90^\circ$

(b)  $180^\circ$

(c)  $270^\circ$

(d)  $360^\circ$

2. An exterior angle of a triangle is equal to: [1]

(a) Sum of its interior opposite angles

(b) Sum of all interior angles

(c) Difference of its interior opposite angles

(d)  $180^\circ$  minus opposite angle

3. Which of the following sets of side lengths can form a valid triangle? [1]

(a) 2 cm, 3 cm, 5 cm

(b) 3 cm, 6 cm, 10 cm

(c) 6 cm, 8 cm, 10 cm

(d) 4 cm, 4 cm, 9 cm

4. The product of a fraction and its reciprocal is always: [1]

(a) 0

(b) 1

(c) Equal to the fraction

(d) Infinite

5. What is the value of  $\frac{3}{7} \times \frac{14}{9}$ ? [1]

(a)  $\frac{1}{3}$

(b)  $\frac{2}{3}$

(c)  $\frac{3}{2}$

(d)  $\frac{7}{3}$

6. Fill in the blank: The meeting point of the medians of a triangle is called the \_\_\_\_\_. [1]

7. Fill in the blank:  $5 \div \frac{1}{3} =$  \_\_\_\_\_. [1]

8. Fill in the blank: An altitude of a triangle forms an angle of \_\_\_\_\_ $^\circ$  with the base. [1]

9. True or False: Every equilateral triangle is also an isosceles triangle. [1]

10. True or False: Dividing a number by  $\frac{1}{2}$  is equivalent to doubling the number. [1]

## SECTION B: VERY SHORT ANSWER QUESTIONS (6 Marks)

11. The angles of a triangle are in the ratio 1 : 2 : 3. Find the measure of all three angles. [2]

12. Find the reciprocal of  $3\frac{2}{5}$ . [2]

13. In  $\triangle ABC$ , exterior angle  $\angle ACD = 110^\circ$ . If interior opposite angle  $\angle B = 45^\circ$ , find  $\angle A$ . [2]

## SECTION C: SHORT ANSWER QUESTIONS (12 Marks)

14. One acute angle of a right-angled triangle is twice the other. Find both acute angles. [3]

15. Solve:  $3\frac{1}{2} - 1\frac{5}{6} + 2\frac{2}{3}$ . [3]

16. The sides of a triangle are 8 cm and 11 cm. Between which two numbers must the length of the third side fall?

[3]

17. A rope of length  $15\frac{3}{4}$  metres is cut into pieces of length  $\frac{3}{4}$  metre each. How many pieces are obtained? [3]

#### SECTION D: LONG ANSWER QUESTIONS (8 Marks)

18. State and prove the Angle Sum Property of a triangle using a parallel line. [4]

19. The area of a rectangular field is  $65\frac{1}{3} \text{ m}^2$ . If its breadth is  $5\frac{1}{3} \text{ m}$ , find its length and perimeter. [4]

#### SECTION E: CASE STUDY / COMPETENCY-BASED QUESTION (4 Marks)

##### 20. Solar Panel Rooftop Truss:

A solar installation crew mounts photovoltaic panels on a rooftop triangular truss structure  $ABC$ :

- Base beam  $BC$  is horizontal.
- Strut  $AB$  has inclination  $\angle B = 35^\circ$  with the horizontal.
- Ridge angle  $\angle A = 85^\circ$ .

(a) Calculate the tilt angle of strut  $AC$  with the horizontal roofline ( $\angle C$ ). [1]

(b) If an exterior bracket at vertex  $B$  measures the exterior angle, what is its value? [1]

(c) If strut  $AB = 4.2 \text{ m}$  and strut  $AC = 2.8 \text{ m}$ , what is the theoretical maximum allowable length for the base beam  $BC$  to keep the truss rigid? [2]